



Letter to the Editor

Evaluation of journal policies to increase promotion of transparency and openness in sport science research On behalf of


Dear Prof Gordon Waddington,
Editor-in-chief, *Journal of Science and Medicine in Sport*

Increased transparency and openness of sport science research will improve the ability to appraise, replicate, and implement research findings. However, sport science journals need to support and prioritise practices that bolster transparency and openness to facilitate this.^{1,2} Journal policies that actively support transparency and openness have been shown to improve the quality, reproducibility and replicability of research.³ To date, over 1100 journals and funding agencies have ascribed to implementing one or more of the eight standards for transparent and open research described in the Transparency and Openness Promotion (TOP) Guidelines.⁴

We report an appraisal of how well the policies of the leading 38 sport science journals support transparent and open research practices. The *Journal of Science and Medicine in Sport* was evaluated using the TOP Factor (<https://osf.io/t2yu5/>) – an alternative metric for evaluating the degree to which journal policies promote transparency and openness,⁵ and the International Committee of Medical Journal Editors (ICMJE) disclosure of interest requirements.⁶ The TOP Factor is calculated as the sum of journal implementation of the eight modular standards within the TOP Guidelines and an additional standard related to publication of replication studies. The requirements for disclosure of conflicts of interest were evaluated using the four standards stated in the ICMJE disclosure form.⁶ The *Journal of Science and Medicine in Sport*'s TOP Factor and conflict of interest scores are provided in Figs. 1 and 2 respectively, alongside summary data from the top 38 sport science journals.

Briefly, the *Journal of Science and Medicine in Sport* scored well for TOP Factor items “design and analysis transparency” and “study preregistration” and poorly for items “data citation”, “code citation”, “data transparency”, “code transparency”, “research materials transparency”, “analysis preregistration”, and “replication”, and had a total TOP Factor score of 4/27 (higher scores represent increased requirement for transparency and openness by the journal). The *Journal of Science and Medicine in Sport* was above the mean TOP Factor score of $2.05 \pm 1.99/27$, and was above the mean ICMJE disclosure of conflicts score ($2.84 \pm 1.62/4$) compared to the top 38 sports science journals.

We strongly encourage the *Journal of Science and Medicine in Sport* to review and, where necessary, update your journal's policies (see <https://cos.io/top/> for advice and resources to implement this). Greater transparency and openness will improve the conduct and translation of sport science research to ultimately move the field forward. Moreover, complete and transparent disclosure of conflicts of interest will increase confidence in the research findings. By publishing this letter, the *Journal of Science and Medicine in Sport* is taking action towards improving the transparency and openness of sports science research.

Availability of data and material

All data is available on OSF (<https://osf.io/d2t4s/>).

Code availability

All code is available on OSF (<https://osf.io/d2t4s/>).

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Declaration of interest statement

All other authors declare that they have no conflicts of interest.

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